ML for Mech Eng (CoEP)

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Outline

About Me

Yogesh Haribhau Kulkarni

Bio:

- 20+ years in CAD/Engineering software development
- Got Bachelors, Masters and Doctoral degrees in Mechanical Engineering (specialization: Geometric Modeling Algorithms).
- Currently doing Coaching in fields such as Data Science, Artificial Intelligence Machine-Deep Learning (ML/DL) and Natural Language Processing (NLP).
- Feel free to follow me at:
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Office Hours: Saturdays, 2 to 5pm (IST); Free-Open to all; email for appointment.

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Naive Bayes

Naive Bayes

What is it ?

- Statistical method for classification.
- Supervised Learning Method.
- Assumes an underlying probabilistic model, the Bayes theorem.
- Can solve problems involving both categorical and continuous valued attributes.
- Named after Thomas Bayes, who proposed the Bayes Theorem.

Bayes Theorem

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Probability (recap)

- Single Probability: "X has the value x" $P(X = x)$
- Joint probability: "X and Y": $P(X = x, Y = y)$
- The probability that variable X takes on the value x and variable Y has the value y: $P(X, Y)$
- Conditional probability: "Y" given observation of "X": $P(Y = y|X = x)$
- If "X" and "Y" are independent then $P(X, Y) = P(X) \times P(Y)$
- For dependent "X" and "Y" : $P(X, Y) = P(Y|X) \times P(X)$

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Probability (recap)

Derivation

$$P(X,Y) = P(Y \mid X) \times P(X)$$

$$P(Y,X) = P(X \mid Y) \times P(Y)$$

$$P(X,Y) = P(Y,X)$$

$$P(X,Y) = P(Y \mid X) \times P(X) = P(X \mid Y) \times P(Y)$$

<u>Bayes' Theorem:</u> $$P(Y \mid X) = \frac{P(X \mid Y) \times P(Y)}{P(X)}$$

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Example

- On a college campus, you meet a guy, say “Tom”. He appears Shy.
- You are asked to guess, whether Tom is a Math PhD student or a Business MBA student. Assume that these are the only two types in that campus.
- Your guess is probably: Math PhD student, assuming that we have seen many Math PhD students, Shy.
- Let us check if this is true, using Bayes theorem.

(Ref: A visual guide to Bayesian thinking)

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Example

- If you multiply corresponding values, $1x75 : 10x15$, it is $75 : 150$, ie $1 : 2$
- So that's the shaded area ratio. Note that Math is in column 1, Business column 2, is 2.
- That's called Posterior Probability.
- So, Tom is more likely to be a Business MBA guy than Math PhD!!

(Ref: A visual guide to Bayesian thinking)

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Bayes Learning

(Ref: Georgia tech, Udacity, Machine Learning)

Bayesian Learning

- Goal of machine learning is: We want best model/function for the given data.
- Can we say? Its equivalent to: We want most "probable" model, for the given data.
- That's represented by: $P(h|D)$ where h is hypothesis (think of it as a function) and D is data.
- We can try many hypotheses, and find max of them.

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Bayes Rule (recall)

- $P(h|D) = \frac{P(D|h)P(h)}{P(D)}$
- Chain Rule: $P(a \cap b) = P(a|b)P(b) = P(b|a)P(a)$ as its symmetric
- Better way to understand is $P(a|b)$ is probability of a given b, meaning within b, how much is a, ie the intersection of a and b.
- So $P(a|b)$ is $\frac{a \cap b}{b}$, right?

De-constructing the formula

- The bottom term, $P(D)$ is called as Prior on the Data, ie belief of seeing the data.
- As we are comparing many hypotheses, and this bottom term being there in all such calculations, one can ignore it. It's a normalization term.
- $P(D|h)$ is like learning backwards. Its easy to see it in the machine learning dataset. Its basically the LIKELIHOOD of seeing Data given the hypothesis.
- Note : $D = \{(x_i, y_i)\}$
- So it means, given x_i, and the hypothesis that we are testing, meaning the function, whats the likelihood of seeing y_i?

Example

- Hypothesis h = return true if $x > 10$
- Data $x = 7$
- Now the question is what is the probability of $y == True$.
- Its 0.
- What for False?
- Its 1.

Back to formula

- As we have labeled data, its easier to compute/adjust $P(D|h)$ (as h and D are known) than $P(h|D)$
- $P(h)$ is prior on h.
- Probability of this hypothesis h occurring amongst all hypotheses.
- Its actually domain knowledge, the features, the type of model we are trying.
- It's the probability of selected features or model.

Properties

- If we have left hand side, ie ,$P(h|D)$ to go up, what should be changed on the right hand side?
- $P(h)$ can go up. Probability of getting good hypothesis, ie better features or model, without seeing data, can go up
- $P(D|h)$: Once you see data, for that data, if you get that nicely predicting with h, then we are good. Good accuracy giving h.
- $P(D)$ cannot change, (ie cannot go down)

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Example

- A man goes to a doctor. He gives him a test for a rare disease. The test generally returns correct positive results 98
- $TP = 0.98$, $TN = 0.97$, $FP = 0.03$, $FN = 0.02$
- The disease in general, in the population is to only .8% people.
- The result: the test comes positive.
- Question: Does he really have the disease?

Solution

- $P(disease = true | test == positive) = 0.98x0.008 = 0.00784$ (ignored denominator term)
- $P(disease = false | test == positive) = 0.03x0.992 = 0.02976$. This is bigger.
- SO, don't Panic!!
- Just within results ie 0.00784 and 0.02976, Probability of disease not being there, is $0.02976/(0.02976 + 0.00784) = 0.79$, ie about 79% probability that you won't have disease even if the test was Positive. Only remaining 21% chance that you may have the disease.

Algorithm

- Good part is, as we want argmax only, we don't need denominator P(D) , which is hard to calculate anyway.
- $P(h)$ is also hard to compute. So if we further drop that and find $argmaxP(D|h)$ than its called Maximum Likelihood Hypothesis.
- That just makes an assumption, that all $P(h)$ are same.
- All model or feature scheme are equally likely

```
For each h in H
    Calculate p(h|D) = P(D|h)P(h)/P(D)
Output:
    H = argmax_h P(h|D)
```

Bayesian Classifier

Bayes Classifier

- A classifier is a function
- It takes features as inputs, so $x_1, x_2 \ldots$
- Gives y as output, which could be $y = 1, y = 2, \ldots$. These are the classes. For binary classifier y can take only $0, 1$.
- x can be real or numeric and it could also be categorical.
- With training data, the function is learnt. That's the classifier model.

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Bayes Classifier

- Bayesian Classifier is probabilistic.
- Meaning, it does not produce y class values directly, but it gives probabilities of $y = 0$ and $y = 1$
- We pick max and say that, that's the output class
- $\hat{y} = argmax P(y|x)$

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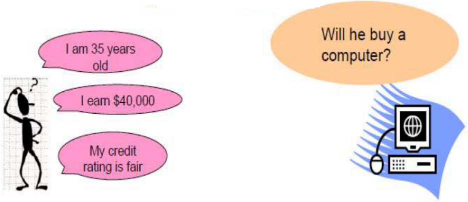
Bayesian classification

Example



- X: Independent Variables: Age, Income, Credit Rating
- Y: Dependent Variable: Will buy computer or not? Binary.

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Bayesian classification

Bayesian probability of a class:

$$P(y|x) = \frac{\overbrace{P(x|y)}^{\text{class model}}\overbrace{P(y)}^{\text{prior}}}{\underbrace{\sum_{y'} P(x|y')P(y')}_{\text{normalizer } P(x)}}$$

- Where denominator term of Bayes Theorem: $P(Y|X) = \frac{P(X|Y)P(Y)}{P(X)}$ is expanded
- Its summation of Probabilities that customer is 35 given that he buys computer, customer has fair credit rating given that he buys computer, and so on. And, also similarly for when he does not buy computer.

Bayesian classification Example

$$P(y|x) = \frac{P(x|y)P(y)}{\sum_{y'} P(x|y')P(y')}$$

Example:
y ... UK patient has Ebola
x ... observed symptoms

- y is binary, whether Ebola is present or not.
- x are many, whether s/he has temperature, looks pale, and similar symptoms.

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Bayesian classification Example

$$P(y|x) = \frac{P(x|y)P(y)}{\sum_{y'} P(x|y')P(y')}$$

Example:
y ... UK patient has Ebola
x ... observed symptoms

- P(y): In general, without considering any features or symptoms, whats the probability of having Ebola.
- Say, in general, probability of having Ebola is very less. Its rare disease in the developed world.
- So, even if there are some symptoms, likelihood of they being for Ebola, is less. That's the role "prior" plays.

Bayesian classification Example

$$P(y|x) = \frac{P(x|y)P(y)}{\sum_{y'} P(x|y')P(y')}$$

Example:
y ... UK patient has Ebola
x ... observed symptoms

- $P(x|y)$: Given that he has Ebola, within this sub population, what are the chances of he having x symptom.
- Gen that he has Ebola how probable is he having (high) temperature.
- Similarly for other symptoms, or Xs

Independence assumption

- Trick is to assume that all these x variables are independent, given y.
- That makes combined probability as just multiplication of individual conditional probabilities.
- So, it becomes sum of multiplications only

$$P(x_1 \ldots x_d | y) = \underbrace{\prod_{i=1}^{d} P(x_i | x_1 \ldots x_{i-1}, y)}_{\text{chain rule (exact)}} = \underbrace{\prod_{i=1}^{d} P(x_i | y)}_{\text{independence}}$$

- Individual probabilities are: whats probability of a particular pixel being black given the digit is 3.
- Multiply for all variables Xs.

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Independence assumption

- Joint probability of Y and many features $X_1, X_2, X_3, \ldots, X_n$ can be expressed as the chain rule:
  $P(Y, X_1, X_2, X_3, \ldots, X_n)$
  $= P(X_1, X_2, X_3, \ldots, X_n | Y) P(Y)$
  $= P(Y) P(X_1 | Y) P(X_2 | Y, X_1) P(X_3 | Y, X_1, X_2) \ldots$
- This is messy, but if we naively assume independence, then
  $P(X_2 | Y, X_1) = P(X_2 | Y)$
  $P(X_3 | Y, X_1, X_2) = P(X_3 | Y)$
  $P(X_n | Y, X_1, X_2, \ldots) = P(X_n | Y)$

Intuition

Without this assumption, estimating the joint probability of many features would need a training example for nearly every possible combination of feature values – practically impossible once you have more than a handful of features. Assuming conditional independence collapses that into one small probability per feature, each estimated separately from the data, which is why the "naive" shortcut is what makes the classifier usable at all.

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Why is Bayes Classifier so popular?

- Look at what it eventually comes down to.
- Just some counting and multiplication.
- We can pre-compute all these terms, and so classifying becomes easy, quick and efficient.

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Naive Bayes Example - Fruits Classification

Example

- So, let's say we have data on 1000 pieces of fruit.
- we know 3 features of each fruit, whether it's long or not, sweet or not and yellow or not
- Outcome is the type: a Banana, Orange or some Other fruit.

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Example

Probability of "Evidence". P(X)

- P(Long) = 0.5
- P(Sweet) = 0.65
- P(Yellow) = 0.8

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Example

Based on our training set we can also say the following:

- From 500 bananas 400 (0.8) are Long, So $P(Long|Banana) = 0.8$
- 350 (0.7) are Sweet. So, $P(Sweet|Banana) = 0.7$
- 450 (0.9) are Yellow, So, $P(Yellow|Banana) = 0.9$

Example

Testing

- Given the features of a piece of fruit and we need to predict the class
- Test fruit is Long, Sweet and Yellow
- The one (fruit) with the highest probability (score) being the winner.

Example

```
P(Banana|Long, Sweet and Yellow)
      P(Long|Banana) * P(Sweet|Banana) * P(Yellow|Banana) * P(banana)
    = _______________________________________________________________
                     P(Long) * P(Sweet) * P(Yellow)

    = 0.8 * 0.7 * 0.9 * 0.5 / P(evidence)

    = 0.252 / P(evidence)


P(Orange|Long, Sweet and Yellow) = 0


P(Other Fruit|Long, Sweet and Yellow)
      P(Long|Other fruit) * P(Sweet|Other fruit) * P(Yellow|Other fruit) * P(Other Fruit)
    = ____________________________________________________________________________________
                                     P(evidence)

    = (100/200 * 150/200 * 50/200 * 200/1000) / P(evidence)

    = 0.01875 / P(evidence)
```

By an overwhelming margin ($0.252 >> 0.01875$), we classify this Sweet/Long/Yellow fruit as likely to be a Banana.

Naive Bayes Example - Loan Default

Example

Tid	Refund	Marital Status	Taxable Income	Evade
1	Yes	Single	125K	No
2	No	Married	100K	No
3	No	Single	70K	No
4	Yes	Married	120K	No
5	No	Divorced	95K	Yes
6	No	Married	60K	No
7	Yes	Divorced	220K	No
8	No	Single	85K	Yes
9	No	Married	75K	No
10	No	Single	90K	Yes

- Target class: Evade
- Predictor variables: Refund, Status, Income
- What is probability of Evade given the values of Refund, Status, Income?
- $P(E|R, S, I)$

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Example

Tid	Refund	Marital Status	Taxable Income	Evade
1	Yes	Single	125K	No
2	No	Married	100K	No
3	No	Single	70K	No
4	Yes	Married	120K	No
5	No	Divorced	95K	Yes
6	No	Married	60K	No
7	Yes	Divorced	220K	No
8	No	Single	85K	Yes
9	No	Married	75K	No
10	No	Single	90K	Yes

- Test instance is:
  - Refund=Yes
  - Status=Married
  - Income=60K
- Issue: we don't have any training example that these same three attributes values.
- How to compute? $P(E|R, S, I)$

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Estimating Prior Probabilities

Class target $P(Y)$ (Count these from the table below)

Tid	Refund	Marital Status	Taxable Income	Evade
1	Yes	Single	125K	No
2	No	Married	100K	No
3	No	Single	70K	No
4	Yes	Married	120K	No
5	No	Divorced	95K	Yes
6	No	Married	60K	No
7	Yes	Divorced	220K	No
8	No	Single	85K	Yes
9	No	Married	75K	No
10	No	Single	90K	Yes

- $P(Evade{=}yes)$
- $= 3/10$
- $P(Evade{=}no)$
- $=7/10$

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Estimating Prior Probabilities

Categorical Attributes $P(X_1|Y)$ (Count these from the table below)

Tid	Refund	Marital Status	Taxable Income	Evade
1	Yes	Single	125K	No
2	No	Married	100K	No
3	No	Single	70K	No
4	Yes	Married	120K	No
5	No	Divorced	95K	Yes
6	No	Married	60K	No
7	Yes	Divorced	220K	No
8	No	Single	85K	Yes
9	No	Married	75K	No
10	No	Single	90K	Yes

- $P(Refund{=}yes \mid Evade{=}no)$
- $= 3/7$
- $P(Status{=}married \mid Evade{=}yes)$
- $=0/3$
  - *Yikes!*
  - *Will handle the 0% probability later*

Estimating Prior Probabilities

Continuous Attributes $P(X_1|Y)$ (Count these from the table below)

Tid	Refund	Marital Status	Taxable Income	Evade
1	Yes	Single	125K	No
2	No	Married	100K	No
3	No	Single	70K	No
4	Yes	Married	120K	No
5	No	Divorced	95K	Yes
6	No	Married	60K	No
7	Yes	Divorced	220K	No
8	No	Single	85K	Yes
9	No	Married	75K	No
10	No	Single	90K	Yes

- For continuous attributes:
  1. Discretize into bins
  2. Two-way split:
     - $(A <= v)$ or $(A > v)$

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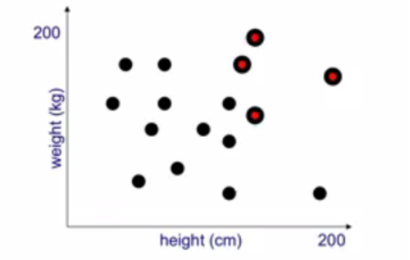
Naive Bayes Classification with Continuous Input Variables

Continuous Variable Example

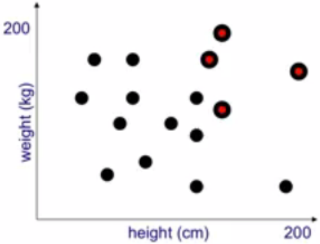
- Features: Height and Weight
- Outcome: class children, adults



Continuous Variable Example

Compute priors, for children and Adults

$$P(a)=\frac{4}{4+12}=0.25\ ;\ P(c)=0.75$$



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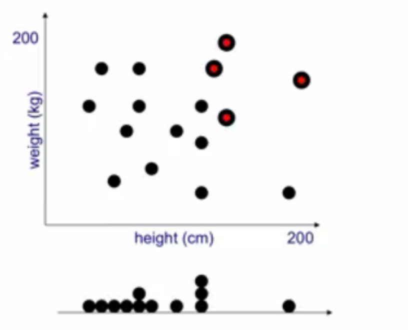
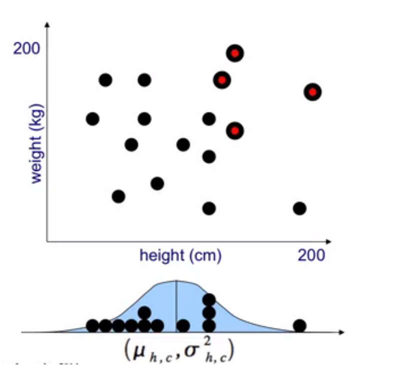
Continuous Variable Example

- Now, within children (black dots), look at only Height for now.
- That would be projection of black dots on X axis.



- One child is really tall.
- Rest are sort of together on smaller values.

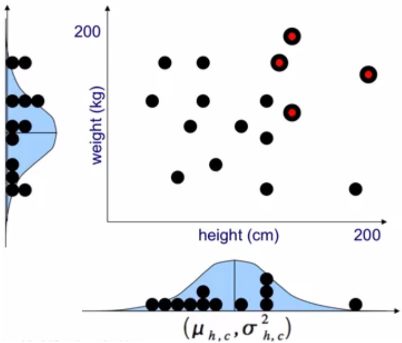
Continuous Variable Example



- Modeling the Histogram with Gaussian (as this is continuous data!!)
- Its not a good fit, but Gaussian is easy to handle.

Continuous Variable Example

- Same thing for Weights.
- These, both, Gaussians are independent.
- Their respective values can be multiplied



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Quick Check: Naive Bayes

Think About It

In the rare-disease example, most of the probability mass ended up on "no disease" even though the test came back positive, despite the test being 97-98% accurate. Why does a low base rate (prior) "defeat" a seemingly accurate test, and what does this imply about interpreting positive results for rare conditions in general?

Quick Check: Naive Bayes

Answer

Because the healthy population is so much larger than the sick population, even a small false-positive rate applied to that huge healthy group produces more false alarms than true positives from the tiny sick group. This is the base-rate fallacy: a test's accuracy percentage alone never tells you the probability of disease given a positive result – you must always weigh it against the prior, exactly what Bayes' Theorem forces you to do.

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Pros and cons

Advantages

- It's relatively simple to understand and build
- It's easily trained, even with a small data-set
- It's fast!
- It's not sensitive to irrelevant features

Disadvantages

- It assumes every feature is independent, which isn't always the case

Naive Bayes (Summary)

- Classification based on Bayes theorem
- Assumption of independence between predictors.
- That is a too 'naive' assumption to make.
- Naive Bayesian model is easy to build and particularly useful for very large data sets.
- Along with simplicity, Naive Bayes is known to outperform even highly sophisticated classification methods.

Naive Bayes (Summary)

- In many datasets, relationship between attributes and class variable is non-deterministic.
- Why?
  - Noisy data
  - Confounding and interaction of factors
  - Relevant variables not included in the data
- Scenario
  - Risk of heart disease based on individual's diet and workout frequency
  - Most people who "work out" and have a healthy diet don't get heart disease. Yet, some healthy individuals still do: Smoking, alcohol abuse

Thanks ...

- Office Hours: Saturdays, 3 to 5 pm (IST); Free-Open to all; email for appointment to yogeshkulkarni at yahoo dot com
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